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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ALAMAR BIOSCIENCES, INC.,
Petitioner,

v.

OLINK PROTEOMICS AB,
Patent Owner.

IPR 2024-01353
Patent 7,883,848 B2

Before SUSAN L. C. MITCHELL, TAWEN CHANG, and TIMOTHY G. MAJORS, *Administrative Patent Judges.*

CHANG, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)
Dismissing Petitioner's Motion to Exclude Evidence as Moot
37 C.F.R. § 42.64

I. INTRODUCTION

This is a Final Written Decision in an inter partes review challenging the patentability of claims 1–20 of U.S. Patent No. 7,883,848 B2 (Ex. 1001, "the ’848 patent"). We have jurisdiction under 35 U.S.C. § 6.

Petitioner has the burden of proving unpatentability of the challenged claims by a preponderance of the evidence. 35 U.S.C. § 316(e). Having reviewed the parties’ arguments and cited evidence, for the reasons discussed below, we find that Petitioner has not demonstrated by a preponderance of the evidence that claims 1–20 are unpatentable. Additionally, for the reasons discussed below, we dismiss Patent Owner’s Motion to Exclude as moot.

A. Procedural History

Petitioner Alamar Biosciences, Inc. filed a Petition requesting *inter partes* review of claims 1–20 of the ’848 patent. Paper 2 ("Pet."). Patent Owner, Olink Proteomics AB, timely filed a Preliminary Response. Paper 6 ("Prelim. Resp."). In view of the then-available preliminary record, we instituted an *inter partes* review. Paper 7 ("Inst. Dec.").

After institution, Patent Owner filed a Response to the Petition. Paper 18 ("PO Resp."). Petitioner filed a Reply. Paper 24 ("Reply"). Patent Owner filed a Sur-reply. Paper 32 ("Sur-reply").

Patent Owner filed a Motion to Exclude certain evidence. Paper 33 ("Mot."). Petitioner opposed that motion. Paper 34 ("Opp."). Patent Owner filed a Reply. Paper 35 ("Mot. Reply").

On December 9, 2025, we held an oral hearing, the transcript of which is of record. Paper 38 ("Tr.").

B. Real Parties-in-Interest

Petitioner identifies itself as the real party-in-interest. Pet. 73. Patent Owner identifies itself, Olink Proteomics Inc., and Thermo Fisher Scientific Inc. as the real parties-in-interest. Paper 4, 2.

C. Related Matters

Petitioner and Patent Owner indicate the ’848 patent is the subject of litigation in *Olink Proteomics AB and Olink Proteomics Inc. v. Alamar Biosciences, Inc.*, C.A. No. 1:23-cv-01303-MN, pending in the United States District Court for the District of Delaware, which was filed on November 15, 2023. Pet. 73; Paper 4, 2.

D. The ’848 Patent and Relevant Background

The ’848 patent is titled “Regulation Analysis by CIS Reactivity, RACR.” Ex. 1001, code (54). The ’848 patent was filed on July 10, 2006, and claims priority to provisional application No. 60/697,415, filed on July 8, 2005. Ex. 1001, codes (22), (60).[1]

The ’848 patent states that, “[t]o identify molecular networks, information concerning at least two molecular species must be gathered since there are at least two molecules involved in any interaction.” Ex. 1001, 1:23–25. The ’848 patent describes “[s]everal . . . approaches . . . in the art which can be utilized for interaction screening,” including protein microarrays, yeast two hybrid system, display techniques such as phage

[1] Petitioner and Patent Owner both assume a July 8, 2005, priority date. Pet. 11–12 (describing the alleged level of skill of a person of ordinary skill in the art “as of July 2005, the earliest possible priority date for the ’848 patent”); PO Resp. 10 (describing alleged level of ordinary skill in the art “before July 8, 2005”). For purposes of this Decision, we apply a July 8, 2005, priority date.

display, DNA templated synthesis (DTS), and proximity ligation. *Id.* at 1:57–5:13.

1. DNA Templated Synthesis (DTS)

The ’848 patent teaches that “[i]n DTS, two molecules are forced together by using nucleic acids to allow the molecules to interact to a higher degree than if they were not brought in proximity,” for instance when “one of the molecules forced together is attached to a ‘template’ nucleic acid which specifically directs the second compound by direct hybridization of the nucleic acid attached to the second molecule.” Ex. 1001, 3:46–53.

The ’848 patent states, however, that in at least one example of a DTS-based method, “the ‘template’ nucleic acid contains both tags used to identify the compounds which are forced together”; thus, “[t]o create interactions between all members of a library with n members with this approach, the number of ‘template’ oligonucleotides which have to be synthesized are in the range of $n^2/2$, since one specific oligonucleotide has to be synthesized for each individual interaction on the library.” *Id.* at 3:53–60. The ’848 patent further notes that, in this DTS approach, “the link between nucleic acid identification tag and the molecules in the interaction depend on correct hybridization of one tag molecule to the template nucleic acid. If any cross hybridization occurs then the link between the tag and the identity of the interaction is unreliable.” *Id.* at 4:5–10.

2. Proximity Ligation

The ’848 patent states that, according to the proximity ligation method,

> targets are detected by utilizing two or more binders, e.g. antibodies, with known affinity to the target. The method is based upon the co-localization of the binder pair in the

> presence of a target. This target brings the binders into proximity, enabling ligation of nucleic acids located on the binder pair. Thus the joining of the nucleic acid becomes elevated in the presence of the target molecule. The nucleic acid can subsequently be quantified and the amount of nucleic acid corresponds to the amount of target.

Ex. 1001, 4:29–38.

The ’848 patent notes that an embodiment of the prior art proximity ligation method “aims at detection of a defined target.” Ex. 1001, 4:39–40. The ’848 patent states that, thus,

> [a]ll the affinities in the system are known and there is always at least one molecule, the target, which does not have a nucleic acid attached to it. Moreover, the [method] utilizes two or more binders which bind their unlabelled analyte pair-wise in a predefined manner. Several pairs might be used in the same reaction but they are always analyzed in a target specific pair wise fashion, not in a combinatorial fashion.

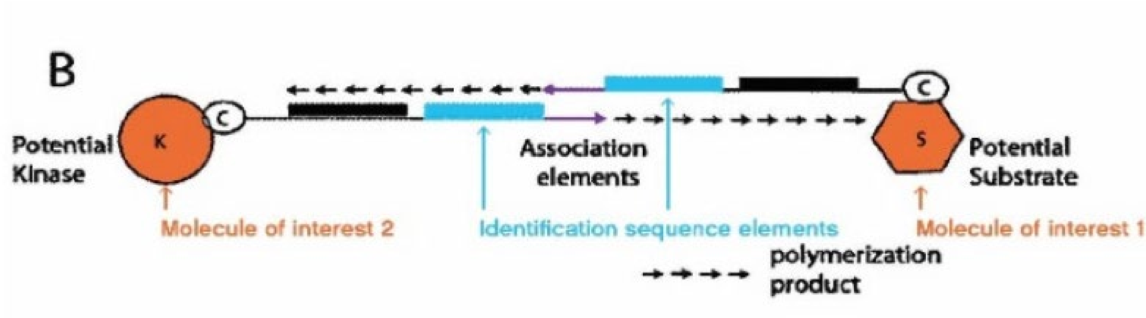
Id. at 4:40–47.

3. The Asserted Invention

The ’848 patent states that its invention is “directed to methods for discovering, as well as detecting molecular interaction networks such as affinity interactions or functional interactions, e.g., enzymes-substrate pairs.” Ex. 1001, 5:22–25.

The ’848 patent teaches tagging “[e]ach molecule of interest (MOI) in [a] library . . . with . . . a nucleic acid moiety (NAM)” comprising “an identification sequence element and an association element,” in order to form “interactors.” Ex. 1001, 5:35–37, 6:7–11. An annotated version of

Fig. 4(B) of the ’848 patent, which appears in the Petition, is reproduced below:



Pet. 1. The annotated Fig. 4(B) provides a schematic depiction of two interactors and is annotated to show the molecules of interest (the potential kinase K and the potential substrate S, in orange), as well as the association elements (in purple) and identification sequence elements (in blue) that together form the NAM. *Id.*

The ’848 patent teaches detecting functional interactions between molecules of interest by binding interactors in proximity to each other based on the affinity between their NAMs to form “cis-reactive cells”; subjecting these cis-reactive cells to conditions that stimulate a desired functional interaction having a detectable trace; selecting the cis-reactive cells exhibiting the trace, and analyzing the oligonucleotides associated with the selected cells to detect the identification sequence elements derived from the NAMs. Ex. 1001, 6:4–23.

The ’848 patent states that, in contrast to the DTS method, in its invention “nucleic acid encoding the tags to identify both members of the interaction are formed upon joining of the two molecules,” such that the number of oligonucleotides required to investigate all interactions in a library of n members is in the order of n, rather than $n^2/2$. Ex. 1001, 3:61–66. The ’848 patent further states that in its invention “the link between the identification sequences in the associated nucleic acid molecule and the phenotype of the molecule is more effectively maintained." *Id.* at 4:11–15. The '848 patent states that its invention differs from proximity ligation in that it does not rely on molecules with predefined affinities. *Id.* at 4:48–5:13.

E. Overview of Challenged Claims

Petitioner challenges claims 1–20 of the '848 patent. Claim 1, reproduced below, is the sole independent claim:

> 1. A method of detecting functional interactions between at least two molecules of interest, the method comprising:
>
> a. forming a plurality of interactors by coupling each molecule of interest with at least one nucleic acid moiety, the nucleic acid moiety comprising an identification sequence element and an association element;
> b. forming a plurality of cis-reactive cells wherein a cis-reactive cell comprises at least two interactors bound in proximity to one another by an associated oligonucleotide formed from the association between at least two nucleic acid moieties, wherein the associated oligonucleotide comprises at least two identification elements derived from the at least two nucleic acid moieties;
> c. subjecting the plurality of cis-reactive cells to conditions which stimulate a desired functional interaction having a detectable trace;
> d. selecting all cis-reactive cells exhibiting the detectable trace; and
> e. subjecting the associated oligonucleotides from the cis-reactive cells selected in step (d) to an analysis that permits detection of the at least two identification sequence elements.

Ex. 1001, 41:37–58.

Claims 2, 3, 19, and 20 recite additional limitations regarding the selection of cis-reactive cells. Ex. 1001, 41:59–67, 44:11–19.

Claim 4 further recites "enhancing the selected associated oligonucleotides by PCR amplification." Ex. 1001, 42:36–39.

Claims 5–10 recite additional limitations regarding the molecules of interest and/or the identification sequence elements. Ex, 1001, 42:40–43:2.

Claim 11 further recites "quantifying the functional interactions." Ex. 1001, 43:3–6.

Claims 12–15 recites additional limitations regarding the analysis permitting detection of the identification sequence elements. Ex. 1001, 43:7–19.

Claim 16 and 17 recite additional limitations regarding the association between NAMs that forms the associated oligonucleotide binding interactors in proximity. Ex. 1001, 43:20–44:3.

Claim 18 recites additional limitations regarding the conditions for stimulating the functional interaction. Ex. 1001, 44:4–10.

F. Asserted Grounds of Unpatentability

Petitioner asserts the following grounds of unpatentability (Pet. 2–3):

Ground	Claims Challenged	35 U.S.C §[2]	Reference(s)/Basis
1	1–15, 17–20	103(a)	Kanan,[3] Neri[4]

[2] The Leahy-Smith America Invents Act ("AIA"), Pub. L. No. 112-29, 125 Stat. 284, 287–88 (2011), amended 35 U.S.C. § 103, effective March 16, 2013. Because the '848 patent has an effective filing date before March 16, 2013, the pre-AIA version of § 103 applies. Our decision, however, would be the same under either version.

[3] Matthew W. Kanan et al., *Reaction Discovery Enabled by DNA-Templated Synthesis and In Vitro Selection*, 431 NATURE 545 (2004) (Ex. 1010, "Kanan" & Ex. 1013, "Kanan Suppl.").

[4] Neri et al., US 2004/0014090 A1, published Jan. 22, 2004 (Ex. 1011, "Neri").

Ground	Claims Challenged	35 U.S.C §[2]	Reference(s)/Basis
2	16	103(a)	Kanan, Neri, Famulok[5]
3	1–8, 11, 12	103(a)	Baez,[6] Landegren[7]
4	14, 15	103(a)	Baez, Landegren, Freskard[8]

Petitioner relies on supporting evidence including the declarations of Norman C. Nelson, Ph.D. (Ex. 1005, "Nelson Declaration" and Ex. 1051, "Second Nelson Declaration"). Patent Owner relies on supporting evidence including the declaration of Dr. Bradley L. Pentelute (Ex. 2008, "Pentelute Declaration").

II. ANALYSIS

A. Principles of Law

"In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (2012) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). This burden of persuasion never shifts to Patent Owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden

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[5] Michael Famulok, *Bringing Picomolar Protein Detection into Proximity*, 20 NATURE BIOTECHNOLOGY 448 (2002) (Ex. 1012, "Famulok").
[6] Baez et al., US 6,511,809 B2, issued Jan. 28, 2003 (Ex. 1003, "Baez").
[7] Landegren et al., WO 01/61037 A1, published Aug. 23, 2001 (Ex. 1004, "Landegren").
[8] Freskgard et al., WO 2005/003778 A2, published Jan. 13, 2005 (Ex. 1009, "Freskgard").

of proof in *inter partes* review). To prevail, Petitioner must demonstrate unpatentability by a preponderance of the evidence. 35 U.S.C. § 316(e).

A claim is unpatentable under 35 U.S.C. § 103 if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *See KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of nonobviousness, if any. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

An obviousness determination requires finding a reason to combine the asserted prior art teachings, accompanied by a reasonable expectation of achieving what is claimed in the challenged patent. *See Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367 (Fed. Cir. 2016). "[A]ny need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed." *KSR*, 550 U.S. at 419–20.

B. Level of Ordinary Skill in the Art

We consider the grounds of unpatentability in view of the understanding of a person of ordinary skill in the art (sometimes referred to herein as "POSA" or "skilled artisan") as of July 8, 2005. *See supra* 3, n.1.

The parties' competing definitions for the level of skill of a person of ordinary skill in the art are set forth below:

Petitioner	Patent Owner
A POSA as of July 2005 . . . would have "(a) an advanced degree, such as a Ph.D., in organic chemistry, molecular biology, biochemistry, or a related field; and at least two years of experience in industry and/or laboratory experience relating to small molecules and proteins and would have been familiar with nucleic acid chemistry and conventional hybridization techniques; and (b) access to one or more team members having an advanced degree, such as a Ph.D., in organic chemistry, molecular biology, biochemistry, or a related field; and at least two years of experience in industry and/or laboratory experience relating to high throughput screening techniques such as nucleic acid sequencing." . . . A person could also have qualified as a POSA with less formal education and more technical or professional experience. Pet. 11–12 (quoting Ex. 1005 ¶ 25).	Before July 8, 2005, a person of ordinary skill in the art (POSA) would have had an advance degree, such as a Ph.D. in chemistry, organic chemistry, biochemistry, biotechnology, molecular biology, or a related discipline, with about 2 years of experience in laboratory research involving small molecule, proteins, nucleic acid chemistry, and hybridization techniques. A POSA would have had familiarity with high throughput assays, such as screening and sequencing techniques. A POSA may also have had less formal education and more years of technical or professional experience. PO Resp. 10 (citing Ex. 2008 ¶¶ 79–80 (citations omitted)).

The primary difference between Patent Owner's and Petitioner's proposals appears to be whether a POSA should have had "familiarity with high throughput assays," or whether they merely needed "access to one or more team members" with such experience. Pet. 11–12; PO Resp. 10.[9]

Neither party has made any arguments that turns on this distinction.[10] Indeed, Dr. Nelson and Dr. Pentelute agree that their definitions of a POSA are similar, and that they would have reached the same conclusions had they applied the definition proposed by the other. Ex. 1051 ¶ 6; Ex. 2008 ¶ 80. Furthermore, describing a person of ordinary skill as part of a "team" is not per se impermissible, so long as testimony relating to issues of obviousness is offered from the vantage point of the ordinarily skilled artisan. *Indivior Inc. v. Dr. Reddy's Labs., S.A.*, 930 F.3d 1325, 1343 (Fed. Cir. 2019) (finding no clear error in the district court's determination that "a person of ordinary skill would principally have a background in pharmaceutical

[9] Petitioner argues that "PO defined a POSA differently from Petitioner by including a person with a Ph.D. in organic chemistry, . . . likely to include PO's expert Dr. Pentelute." Reply 3. However, Petitioner's proposal of a person of ordinary skill in the art also includes a person with a Ph.D. in organic chemistry, so long as they also meet the other requirements set forth in the definition. Pet. 11–12.

[10] Petitioner argues that Dr. Pentelute did not qualify as POSA at the relevant time because his Ph.D. is from 2008, and he lacks pre-2006 publications." *Id.* (citing Ex. 1048, 35:15–19, 185:6–9; Ex. 2010). Patent Owner asserts that Petitioner's allegation is "legally and factually wrong." Sur-reply 11, n.2. We note that Petitioner did not move to strike any portion of Dr. Pentelute's opinion. Furthermore, even assuming Dr. Pentelute was not a POSA in July 2005 as defined above, an expert need not be a person of ordinary skill in the art at the time of the invention to offer expert testimony from the vantage point of a skilled artisan. *Osseo Imaging, LLC v. Planmeca USA Inc.*, 116 F.4th 1335, 1341 (Fed. Cir. 2024) (holding that "an expert can acquire the necessary skill level later and develop an understanding of what a person of ordinary skill knew at the time of the invention").

science or chemistry but 'would be on a team with an engineer or scientist with one to three years of relevant experience'").

Accordingly, for purposes of this decision we adopt Petitioner's proposal of a person of ordinary skill in the art, with the following changes noted in italics below:

> A person with "an advanced degree, such as a Ph.D., in organic chemistry, molecular biology, biochemistry, or a related field; and *about* two years of experience in industry and/or laboratory experience relating to small molecules and proteins and would have been familiar with nucleic acid chemistry and conventional hybridization techniques; and (b) access to one or more team members having an advanced degree, such as a Ph.D., in organic chemistry, molecular biology, biochemistry, or a related field; and *about* two years of experience in industry and/or laboratory experience relating to high throughput screening techniques such as nucleic acid sequencing. A person could also have qualified as a POSA with less formal education and more technical or professional experience.[11]

We emphasize, however, that our decision would not change if we instead used Patent Owner's proposed definition of a POSA.

C. Claim Construction

In AIA proceedings we construe claims "using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. [§] 282(b)." 37 C.F.R. § 42.100(b). Therefore, we

[11] As noted in our Institution Decision, which otherwise applied Petitioner's then-uncontested proposal for the level of ordinary skill, the open-ended language "at least" expands the range of experience indefinitely with no upper bound. Inst. Dec. 10–11. We therefore did not apply that aspect of Petitioner's definition in our Institution Decision, *id.*, and do not apply it here.

construe the challenged claims under the framework set forth in *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–19 (Fed. Cir. 2005) (en banc). Under this framework, claim terms are given their ordinary and customary meaning, as would be understood by a person of ordinary skill in the art, at the time of the invention, in light of the language of the claims, the specification, and the prosecution history of record. *Id*.

Petitioner states that, "[f]or Grounds 1 and 2, the claims are assumed to be given their plain and ordinary meaning based on the understanding of a POSA" and that, "[f]or Grounds 3 and 4, the claim terms are given meaning based on PO's application of the claims in related litigation with Petitioner." Pet. 12–13 (citing Exs. 1007, 1008, 1015).

Patent Owner did not address claim construction in the Preliminary Response, except to argue that the claim construction analysis in the Petition is deficient under 37 C.F.R. § 42.104(b)(3). Prelim. Resp. 11–14.

In our Institution Decision, we preliminarily determined, based on the record existing at the time, that it was unnecessary to more explicitly construe any claim term to decide whether Grounds 1 and 2 satisfies the "reasonable likelihood" standard for instituting trial. Inst. Dec. 12–13. Because we determined that Petitioner had met its burden under 37 C.F.R. § 42.104(b)(3) for Grounds 1 and 2 and also demonstrated a reasonable likelihood that at least one challenged claim of the '848 patent is unpatentable based on those grounds, we did not separately decide the sufficiency of Grounds 3 and 4 under 37 C.F.R. § 42.104(b)(3) in our Institution Decision.

In its Response after institution, Patent Owner asserts that "[t]he '848 patent provides specific descriptions and definitions of several claim terms" consistent with the ordinary and customary meanings of these terms.

PO Resp. 8. Patent Owner's proposed constructions for these claim terms are reproduced below:

Claim Term	PO Proposed Construction
molecules of interest (MOI)	"a molecule about which interactive information is desired" (PO Resp. 8 (quoting Ex. 1001, 8:53–62))
nucleic acid moiety (NAM) comprising an identification sequence element and an association element	a moiety comprising "at least one unique identification nucleic acid sequence so that detection of that sequence infers involvement of the coupled MOI" and "at least one element which enables association with other NAMs" (PO Resp. 8 (quoting Ex. 1001, 8:41–52))
interactor	"at least one MOI coupled with at least one NAM" (PO Resp. 9 (quoting Ex. 1001, 9:33–34))
cis-reactive cell (CRC)	"an association of at least two interactor moieties, joined by an associated oligonucleotide" (PO Resp. 9 (quoting Ex. 1001, 9:35-37))
selecting all cis-reactive cells	"[t]o detect functional interactions in a pool of CRCs the CRCs which display functional interactions *must be selected* for detection and/or analysis" (PO Resp. 9 (quoting ex. 1001, 19:41–43 with emphasis added)) "selecting all of the molecules meeting the defined criteria for CRCs and exhibiting the detectable trace" (Sur-reply 8 (citing Ex. 2008 ¶¶ 94, 98–119))

Patent Owner asserts that "[c]laim terms not addressed above should also be given their ordinary and customary meaning." PO Resp. 10.

In response, Petitioner did not advance additional explicit constructions (other than "plain and ordinary meaning") but argues that

Patent Owner's construction of the term "selecting all cis-reactive cells" "ignores that 'selecting' can be a process, consistent with the dependent claims and the specification's disclosure of 'selecting' procedures." Reply 5. During the hearing, Petitioner's counsel affirmed that "the plain and ordinary meaning is what's been provided" but clarified as follows:

> [W]hen we're looking at what the plain and ordinary meaning of selection is, . . . [t]here can be a process that's involved in it. There can be a step of identifying it . . . .
>
> The ultimate goal . . . is a method of detecting functional interactions between at least two molecules of interest. And so, in Step D, you have to select all cis-reactive cells exhibiting the detectable trace, and then Step E is subjecting the associated oligonucleotides from the cis-reactive cells selected in Step D to an analysis that permits detection of the at least two identification sequence elements.
>
> So, when we're looking at selecting, the goal is to be able to, like in the Kanan workflow, to be able to see those two identification sequences so that you can identify it.

Tr. 57:1–19.

Because the parties have placed the limitation "selecting all cis-reactive cells exhibiting the detectable trace" into controversy, we construe the term below, but only to the extent necessary to resolve the controversy. *See Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1375 (Fed. Cir. 2019) ("The Board is required to construe 'only those terms . . . that are in controversy, and only to the extent necessary to resolve the controversy.'") (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

1. "Selecting all cis-reactive cells exhibiting the detectable trace"

For the reasons discussed below, we agree with Patent Owner that the plain and ordinary meaning of this phrase, in light of the language of the claims, the Specification, and the prosecution history of record, requires "selecting all of the [molecular complexes or structures] meeting the defined criteria for [cis-reactive cells (CRCs)] and exhibiting the detectable trace." Sur-reply 8 (citing Ex. 2008 ¶¶ 94, 98–119). We further agree that cis-reactive cells should be construed to mean "an association of at least two interactor moieties, joined by an associated oligonucleotide." (PO Resp. 9 (quoting Ex. 1001, 9:35-37)). Accordingly, we determine that, properly construed, the claims require that what is "selected" at step 1(d) — what is differentiated or set apart from other molecular complexes or structures in the reaction mixture at the conclusion of step 1(d) — to be "an association of at least two interactor moieties joined by an associated oligonucleotide" that exhibits the detectable trace.

We begin with "the context in which a term is used in the asserted claim," which can be "highly instructive." *Phillips*, 415 F.3d at 1314.

In this case, claim 1 recites, among other things, "forming a plurality of cis-reactive cells wherein a cis-reactive cell comprises at least two interactors bound in proximity to one another by an associated oligonucleotide." Ex. 1001, 41:4–49. The claim then recites "subjecting these formed cis-reactive cells to conditions which stimulate a desired functional interaction having a detectable trace"; "selecting all cis-reactive cells exhibiting the detectable trace"; and then "subjecting the associated oligonucleotides from the cis-reactive cells selected in step (d) to an analysis." *Id.* at 50–58.

The above claim language strongly implies that, although it is the associated oligonucleotides that are subjected to analysis, it is the larger molecular complexes, i.e., the cis-reactive cells as defined in the Specification and formed in step 1(b), that are selected based on the plain language of the claim. *See also* Sur-Reply 7–8 (arguing that "the claim language itself 'expressly defines CRCs with specific criteria 'compris[ing] *at least two interactors* bound in proximity to one another by *an associated oligonucleotide formed from the association between at least two nucleic acid moieties*'").

The claim language likewise suggests that the cis-reactive cells retain their basic structure (i.e., "an association of at least two interactor moieties joined by an associated oligonucleotide"), because step 1(e), subsequent to step 1(d), requires analysis of the associated oligonucleotides *from* the selected cis-reactive cells. Thus, the claim language does not merely require that a method ultimately allows the identification sequences in the cis-reactive cells of interest to be detected or determined; the method must do so by first selecting the cis-reactive cells exhibiting the detectable trace indicative of a functional interaction.

Dependent claims 2, 3, and 19 further recite that "selecting" may be accomplished via "selective separation or selective detection *of the cis-reactive cells*," including by "labeling *the cis-reactive cells* with an affinity agent, or via "affinity purification." Ex. 1001, 41:59–67, 44:11–14 (emphasis added). These dependent claims also support the construction that it is the cis-reactive cells, rather than merely some portion thereof, that are selected, whether by separation or detection.

We next turn to the Specification, which "is always highly relevant to the claim construction analysis." *Phillips*, 415 F.3d at 1315 (quoting

Vitronics Corp. v. Conceptronic, Inc., 90 F.3d 1576, 1582 (Fed. Cir. 1996). The Specification defines a "cis reactive cell" as "an association of at least two interactor moieties, joined by an associated oligonucleotide." Ex. 1001, 9:35–38. *See Phillips*, 415 F.3d at 1316 (explaining that "the specification may reveal a special definition given to a claim term by the patentee," in which case "the inventor's lexicography governs"). This definition is consistent with the claims that recite forming cis-reactive cells that "comprise[] at least two interactors bound in proximity to one another by an associated nucleotide," and further supports the construction of step 1(d) as selecting for the "association of at least two interactor moieties" that are "joined by an associated oligonucleotide" and that exhibits the detectable trace indicative of the desired functional interaction.

A comparison of the two main embodiments disclosed in the Specification provides further support that what must be "selected" in step 1(d) is "an association of at least two interactor moieties, joined by an associated oligonucleotide."

In particular, the embodiment directed to "a method of detecting an *affinity* interaction between at least two molecules of interest" recites a step of "selecting the plurality of associated oligonucleotide" that comprises "at least two identification sequence elements derived from the at least two nucleic acid moieties of two interactors." Ex. 1001, 5:50–53. This is in contrast to the claimed embodiment of a method of detecting *functional* interaction between at least two molecules of interest, in which what is selected is "all cis-reactive cells exhibiting the detectable trace" rather than the associated oligonucleotide.[12] *Id.* at 6:4–23. Although in both cases the associated oligonucleotide will be analyzed to detect the at least two sequence elements identifying the interacting molecules of interest, the Specification distinguishes between what is *selected* in the two embodiments — i.e., cis-reactive cells in methods of detecting functional interactions, versus associated oligonucleotide in methods of detecting affinity interactions. *See also* Sur-reply 9 (discussing similar distinction in the original claims directed to methods of detection affinity and functional interactions).

Petitioner argues that Patent Owner "improperly construes 'selecting' as a single capture event" and "ignores that 'selecting' can be a process, consistent with the dependent claims and the specification of 'selecting' procedures." Reply 5–6.

Petitioner first argues the dependent claims indicate that "'selecting' may occur in different procedures, such as . . . 'selective separation,'

[12] Although functional interactions may include affinity interactions, they also include other types of interactions. *Compare* Ex. 1001, 14:48–51 (defining affinity interactions as "interactions between two or more molecules that induce a non-random distribution of these molecules in a solution comprising these molecules), *with id.* at 15:31–36 (defining functional interactions as "interactions between molecules where the nature of at least one of the partners which participate in the interaction deviates from the nature of that partner when the other participant(s) of the functional interaction is/are present in conditions permitting the functional interaction" and noting that "[f]unctional interactions may include an affinity interaction"). Thus, we do not consider methods of detecting affinity and functional interactions to necessarily be equivalent, particular in view of the Specification and prosecution history, which describes the two methods as distinct embodiments.

'selective detection,' 'affinity separation,' or allowing for 'modification [that] permits the selective detection." Reply 6.

We are not persuaded.[13] A construction requiring the molecular complex or structure selected in step 1(d) to be (1) a CRC (i.e., an association of at least two interactor moieties joined by an associated oligonucleotide) and (2) exhibiting the detectable trace does not preclude the use of any of the methods enumerated in the dependent claims or the Specification (e.g., "selective separation," "selective detection," "affinity purification," "labeling the cis-reactive cells with an affinity reagent," and "introducing a modified molecule wherein the modification permits the selective detection").

In particular, the Specification states:

> As used herein, a "filter" or a "filtering process" is the means employed *to selectively separate* the desired CRCs. Nonlimiting examples of filters for functional interactions include: *affinity purification* of the form which has undergone alteration in the functional interaction; detection of the CRCs by *labelling with affinity reagents which detect interactors* which are included in the functional interaction; and introduction of a *modified molecule which becomes incorporated by the functional interaction* where the modification permits selective detection. Downstream of the filtering process of CRCs

[13] To the extent Petitioner's arguments are directed not to claim construction per se, but to how Patent Owner applies the construction in responding to Petitioner's grounds of invalidity, we address the arguments in Section II.D.3 below. *Cf. Biotec Bologische Naturverpackungen GmbH & Co. KG v. Biocorp, Inc.*, 249 F.3d 1341, 1349 (Fed. Cir. 2001) (determining that, in a method including a "melting" step, "[t]he issue in dispute was the application of the melting step in the accused process, a factual question").

> the compositions of associated NAMs are detected and/or quantified.

Ex. 1001, 19:46–57 (emphases added).

This description of selection supports our construction of the "selecting" step as directed to CRCs, rather than some alternative molecular complex or structure that may be derived from CRCs, but that do not meet the Specification's definition of CRCs. First, the description makes clear that it is the CRCs that are selected and that the detection of associated NAMs occurs only "downstream" of the filtering process, i.e., after the CRCs have been selected. This contrasts with Petitioner's apparent argument that step 1(d) is met if it ultimately enables a determination of the nucleic acid sequences identifying molecules of interest involved in the functional interaction of interest. Tr. 57:1–19 ("So, when we are looking at selecting, the goal is to be able to, like in the Kanan workflow, to be able to see those two identification sequences so that you can identify it.").

Second, the description makes clear that construing the limitation to require selection of the CRCs, as that term is defined in the Specification, is not inconsistent with the various methods of selection recited in the dependent claims or in the Specification. For instance, the Specification explains that, as non-limiting examples, affinity purification would be suitable for functional interactions that result in a structural change to one of the interaction partners (i.e., molecules of interest), including, e.g., adding or removing molecules or parts from an interaction partner, where there are specific affinity reagents for the added, removed, or altered portions. Ex. 1001, 20:10–19. Such affinity purification suggests selection of the CRC as a whole (i.e., "an association of at least two interactor moieties, joined by an associated oligonucleotide"), because the affinity reagent may be for any portion of the molecules of interest altered by the functional interaction.

Likewise, although the Specification explains that conditional detection of the NAMs could also be used to filter (i.e., selectively separate) the desired CRCs, Ex. 1001, 20:25–27, the Specification does not suggest that such conditional detection is of the NAMs other than as part of a CRC. *See, e.g., id.* at 20:27–30 (explaining that, "[i]f the detection output is coupled to the result of the interaction, for example, with an affinity reagent for the modification, only *the CRCs* with positive interaction pairs will be detected") (emphasis added).

The Specification provides another example of conditional detection where the CRCs are hybridized to a DNA oligonucleotide tag array where all possible combinations of identification sequences in the library are present and spatially separated, which is then stained with a labelled affinity reagent specific for the interaction. Ex. 1001, 20:31–36. The Specification notes explicitly that "[t]his approach requires that the MOIs are present during detection," which again supports a construction that construing step 1(d) requires selection *of the CRC* is consistent with the selective detection recited in dependent claim 2.

Petitioner argues that the above embodiment shows that "selecting" may include "multiple procedures — first, hybridizing to a DNA oligonucleotide tag array and, second, staining with a labelled affinity reagent." Reply Br. 6. Petitioner contends that Patent Owner's construction of the phrase further improperly excludes embodiments disclosed in the Specification, because it precludes modification of CRCs in the process of selection. *Id.* at 2, 8 (citing Ex. 1001, 19:3–20:46, 30:11–41, 30:56–67). Petitioner argues that, for instance, the Specification describes "selecting" as including “selective amplification of the different NAM combination,” which “unbinds the interactors during the process,” as well as “remov[al of] molecules or parts from one of the interaction partners.” *Id.* at 8 (citing Ex. 1048, 88:17–22, Ex. 1051 ¶¶ 31–33, Ex. 1001, 19:43–46, 20:10–16).

However, requiring selection of “an association of at least two interactor moieties joined by an associated oligonucleotide” (that exhibits the detectable trace) is not inconsistent with either a multi-step selection process or a process that results in modification of the CRC. Nothing in such a construction requires the CRC to remain unmodified or requires the selection to be a single step rather than a process, so long as what is ultimately “selected” is “an association of at least two interactor moieties joined by an associated oligonucleotide” that exhibits the detectable trace. Indeed, the Specification defines functional interaction as one in which “the nature of at least one of the partners which participate in the interaction deviates from the nature of that partner when the other participant(s) of the functional interaction is/are present in conditions permitting the functional interaction.” Ex. 1001, 15:31–36. Because the claims require subjecting the CRCs in step 1(c) to conditions that stimulate the desired functional interaction having a detectable trace, the claim itself suggests that the CRCs are modified (e.g., with respect to the nature of at least one of the interaction partners and/or the incorporation of the detectable trace).

Neither are we persuaded by Petitioner’s other arguments. Petitioner relies on Dr. Pentelute’s deposition testimony to suggest that “‘selecting’ includes ‘selective amplification of the different NAM combination[s].’” Reply 8 (citing Ex. 1048, 88:17–22). However, in the portion of the transcript cited, Dr. Pentelute is discussing Figure 1, which is a “[s]chematic description of one embodiment for affinity interaction detection." Ex. 1001, 6:55–67; *see* Ex. 1048, 88:17–22.

As previously noted, the Specification distinguishes between *what is selected* in methods for detecting functional interactions and affinity interactions. For example, Figure 2, which describes an embodiment for functional interaction detection, describes selection as being "carried out with respect to at least one detectable alteration introduced by the functional interaction," before "the selected *CRCs* are amplified." Ex. 1001, 7:1–16 (emphasis added).

Petitioner further cites to Example 5 of the Specification and Dr. Nelson's discussion of the example to support its claim construction argument. We agree with Patent Owner, however, that Example 5 is also an embodiment of the method of detecting affinity interaction rather than a functional interaction. Sur-reply 10–11. As Patent Owner points out, the Specification explains that the reaction detected in Example 5 (thiol dimerization) may be viewed as "formation of an affinity interaction with infinite affinity." Sur-reply 10 (citing Ex. 1001, 30:7–8). Although Example 5 states that thiol dimerization may also be compared to "a reaction screen where reactive compounds are identified," in the Example, dimerization (i.e., the reaction) occurs first, before the NAMs of the interacting MOIs are associated by ligation. Ex. 1001, 30:12–23; *see also* Sur-reply 10. This is consistent with the Specification's description of an affinity interaction rather than functional interaction, in which CRCs are formed from the affinity between the NAMs before the functional interaction occurs. *Compare* Ex. 1001, 14:57–60, *with id.* at 15:11–30.

During oral argument, Petitioner's counsel argued that, "while Patent Owner points to other portions of the specification that talk about whether it's affinity interactions or other types of interactions that take place that are ultimately selected, the word selecting is not – the plain and ordinary meaning of selecting does not change depending on . . . what you are selecting for." Tr. 55:15–19. We disagree because the term "selecting" cannot be read in a vacuum divorced from the context of the claim. In the context of Petitioner's arguments that Patent Owner's proposed claim construction excludes embodiments disclosed in the Specification, it is indeed relevant whether the embodiments allegedly excluded are, in fact, claimed.

Finally, Petitioner argues that Dr. Pentelute concedes that the step of selecting the desired CRCs may be a "***process***" that "can be ***modified*** according to the nature of the functional interaction that is being investigated," and that the Specification "provides 'many, many possibilities' for selection." Reply Br. 6–7 (emphasis original) (citing, *e.g.*, Ex. 2008 ¶ 35; Ex. 1048, 73:11–14, 136:23–138:15, 145:7–146:3). Once again, however, we are not persuaded that Petitioner has pointed to an example in the Specification where what was selected, in a method for detecting functional interaction, was a molecular complex or structure other than CRCs, or where modification was one that altered the basic structure of CRCs as "an association of at least two interactor moieties joined by an associated oligonucleotide."

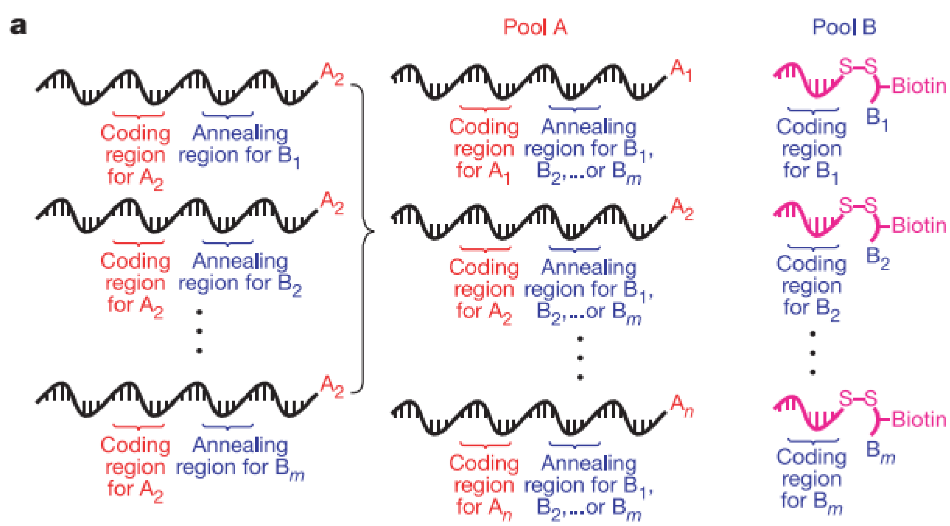
2. Conclusion

Accordingly, we construe the claim phrase "selecting all cis-reactive cells exhibiting the detectable trace" to mean "selecting all associations of at least two interactor moieties joined by an associated oligonucleotide and exhibiting the detectable trace," where an associated oligonucleotide is formed from the combination of two NAMs.

D. Ground 1: Alleged Obviousness over Kanan and Neri

1. Overview of Kanan (Ex. 1010)

Kanan teaches a "reaction discovery approach that uses DNA-templated organic synthesis and *in vitro* selection to simultaneously evaluate many combinations of different substrates for bond-forming reactions in a single solution." Ex. 1010, 545 (endnotes omitted). In this approach, "two pools of DNA-linked substrates [are prepared], with *n* substrates in pool A and *m* substrates in pool B," as shown in Kanan's Figure 1a reproduced below:



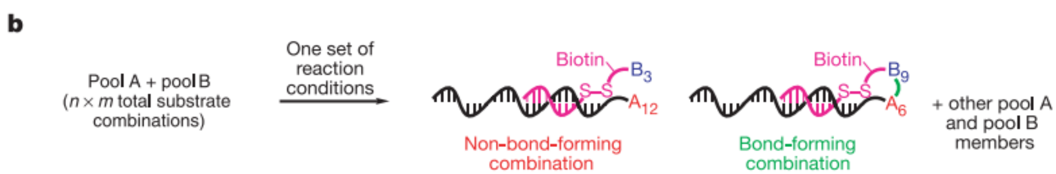
Id. at 545, 546, Fig. 1. Figure 1(a) depicts "[t]wo pools of DNA-linked organic functional groups that associate each of *n* x *m* substrate combinations with a unique DNA sequence." *Id.* at 546, Fig. 1. More particularly,

> [e]ach substrate in pool A is covalently linked to the 5' end of a set of DNA oligonucleotides containing one 'coding region' (uniquely identifying that substrate) and one of *m* different 'annealing regions' (Fig. la). Each of the *m* substrates in pool B is attached to the 3' end of an oligonucleotide containing a coding region that uniquely

> identifies the substrate and complements one of the *m* annealing regions in pool A.

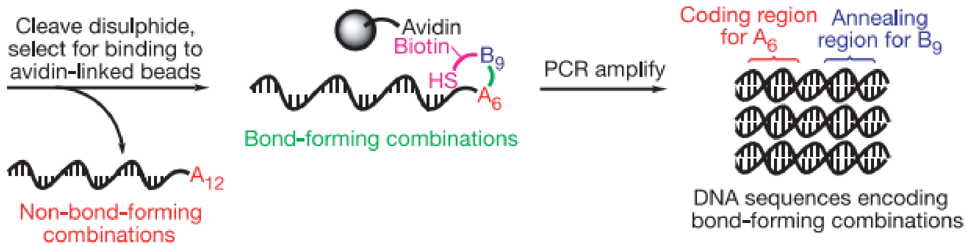
Id. at 545.

In Kanan's approach, "[w]hen pools A and B are combined in a single aqueous solution . . . , Watson-Crick base pairings [among DNA oligonucleotides] organizes the mixture into *n* x *m* discrete pairs of substrates attached to complementary sequences," where "substrates linked to non-complementary oligonucleotides . . . do not react with each other at a significant rate." Ex. 1010, 545. Kanan further teaches "covalently link[ing] each pool B substrate to its corresponding oligonucleotide by a linker containing a biotin group and a disulfide bond [S–S]," as shown in the portion of Figure 1(b) reproduced below:



Id. at 545, 546, Fig. 1. The excerpted portion of Figure 1(b) depicts two pool A substrates (A_{12}, A_6) paired with two pool B substrates (B_3, B_9) through their linked complementary oligonucleotides, where A_{12} and B_3 do not form a bond and A_6 and B_9 do form a bond. *Id.* at 546, Fig. 1.

As Kanan explains and as shown in another portion of Figure 1(b) reproduced below, "[a]fter incubation under a set of chosen reaction conditions, followed by cleavage of the disulphide [*sic*, disulfide] bonds, only pool A sequences encoding bond formation between a pool A and pool B substrate remain covalently linked to biotin":



Ex. 1010, 545, 546, Fig. 1. The excerpted portion of Figure 1(b) depicts the results of cleaving the disulfide bond linking pool B substrates with its oligonucleotide, wherein the only oligonucleotide that remains linked to biotin are those linked to pool A substrates that form a bond with pool B substrates. *Id.* at 545, 546, Fig. 1. Finally, Kanan teaches that "[s]treptavidin affinity selection of the resulting solution separates the biotinylated from non-biotinylated sequences," which allows identification of reactive substrate pairs by PCR amplification of the selected sequences followed by DNA microarray analysis. *Id.* at 545.

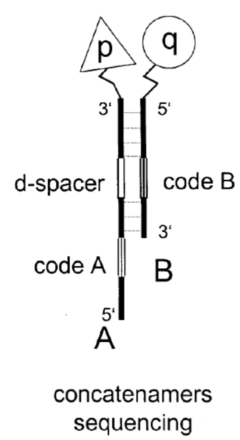
Kanan teaches that its "selection-based approach does not depend on any specific substrate or product property but instead can identify all substrate pairs capable of forming a covalent bond under the reaction conditions." Ex. 1010, 545.

2. Overview of Neri (Ex. 1011)

Neri teaches that "[t]he isolation of specific binding molecules (e.g. organic molecules) is a central problem in chemistry, biology, and pharmaceutical sciences," for instance because "the isolation of specific binders against a relevant biological target typically represents the starting point in the process which leads to a new drug." Ex. 1011 ¶¶ 3, 6. Neri teaches that "[t]ypically, millions of molecules have to be screened, in order to find a suitable candidate," but "[t]he preparation of very large libraries of organic molecules is . . . cumbersome," and "the complexity associated with the identification of specific binding molecules from a pool of candidates grows with the size of the chemical library to be screened." *Id.* ¶ 3.

As a solution, Neri teaches the use of "self-assembling libraries of organic molecules . . . , in which the organic molecules are linked to an oligonucleotide which mediates the self-assembly of the library and/or provide a code associated to each binding moiety." Ex. 1011 ¶ 5. Figure 5 of Neri, partially reproduced below, depicts an example of an assembly of two such organic molecules:

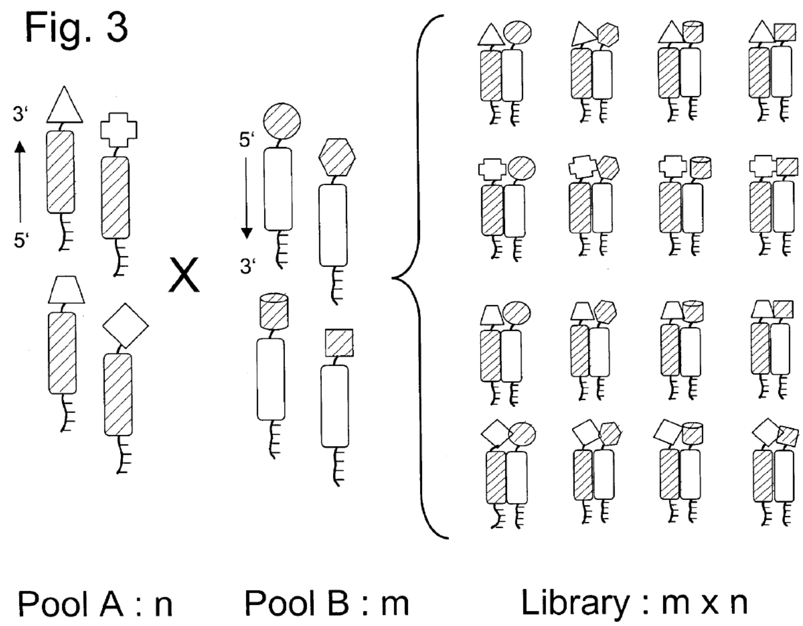
Fig. 5



Ex. 1011, Fig. 5. The reproduced portion of Figure 5 shows oligonucleotides of sub-libraries A and B bearing chemical entity p and q, respectively, at their 3' and 5' ends. *Id.* ¶ 29. Neri explains that "[t]owards the 3' extremity, the DNA sequence [of the oligonucleotide of sub-library A] is designed to hybridize to the DNA sequences at the 5' extremity of oligonucleotides of sub-library B." *Id.* The portions of the oligonucleotides labelled "code A" and "code B" contains a "unique sequence" for each member of sub-library A and sub-library B, respectively. *Id.*

Neri explains that its "encoded self-assembled chemical library (ESACHEL) can be very large (as it originates from the combinatorial self-assembly of two smaller libraries)." Ex. 1001 ¶ 20; *see also id.* ¶ 5. Neri's Figure 3, reproduced below, provides a schematic depiction of an example ESACHEL:

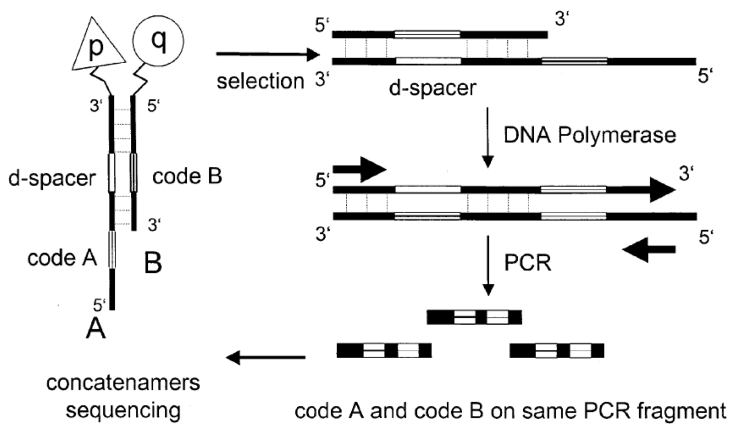
Fig. 3



Id. at Fig. 3. Figure 3 shows n chemical compound-oligonucleotide conjugates in pool A that "self assembles" with m chemical compound-oligonucleotide conjugates in pool B to produce a number of m x n combinations. *Id.* ¶ 27. Neri teaches that "[t]he oligonucleotides of pool A are designed to have" "one portion of the DNA sequence which can hybridize to compounds of pool B" and "a distinctive DNA sequence for each of the n members of [p]ool A." Ex. 1011 ¶¶ 48–50. The oligonucleotides of pool B are similarly designed to have "one portion of the DNA sequence which can hybridize to compounds of pool A" and "a distinctive DNA sequence for each of the m members of [p]ool B." *Id.* ¶¶ 53–55. Neri teaches that the identities of binders of interest can be obtained after capture using the code associated with each binding moiety. *Id.* ¶ 20; *see also id.* ¶ 5.

Neri teaches that, "[a]fter the capture of the desired binding specificities on the target of interest, the 'binding code' can be decoded by a number of experimental techniques, including a modified polymerase chain reaction (PCR) technique followed by sequencing. Ex. 1011 ¶ 5. In one such decoding strategy, illustrated in Neri's Figure 5 reproduced below, PCR fragments are generated that carry the code of pairs of sub-library members whose combinations had the desired binding specificities on the target of interest, which allows the corresponding heterodimerized chemical entities to be identified. *Id.* ¶ 112.

Fig. 5



As shown above in Figure 5, "[o]ligonucleotides of sub-library B remain stably annealed to oligonucleotides of sub-library A, and can work as primers for a DNA polymerase reaction on the template A." Ex. 1011 ¶ 117. Neri explains that the DNA segment resulting from this polymerase reaction, which carries both code A and code B," can then be "amplified (typically by

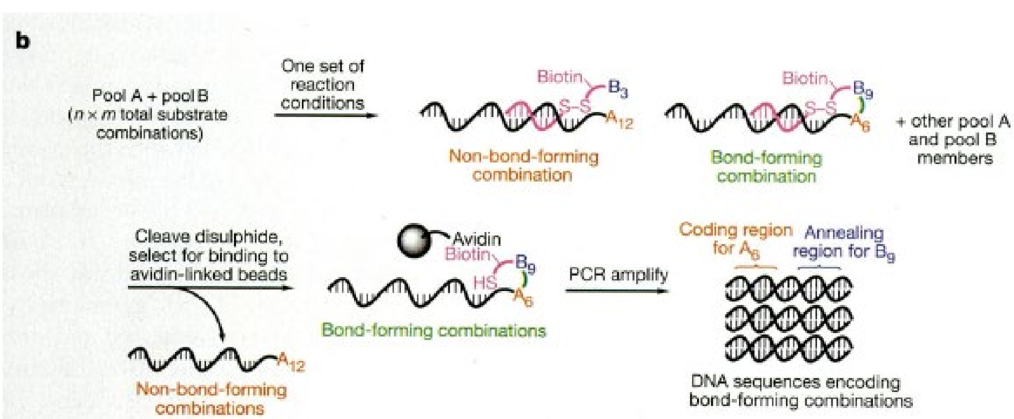
PCR) using primers which hybridize at the constant extremities of the DNA segment." *Id.*

3. Analysis of Claim 1

The parties dispute whether the combination of Kanan and Neri teaches or suggests claim elements 1(d) and 1(e), and whether an ordinarily skilled artisan would have had reason to combine Kanan and Neri to arrive at the claimed invention, with a reasonable expectation of success. PO Resp. 1–2, 20–57; Sur-reply 2.

a. "Selecting all cis-reactive cells exhibiting the detectable trace"

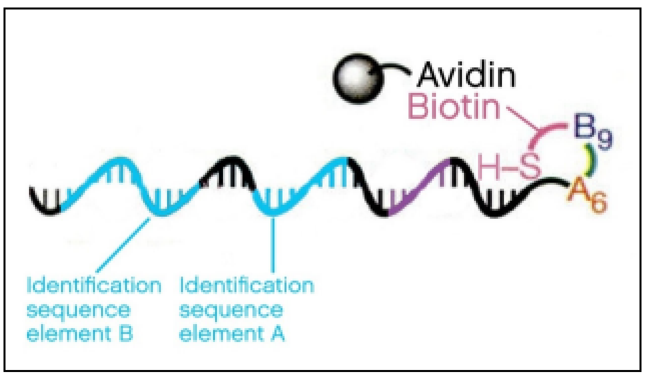
As discussed above, Kanan teaches that each pool B substrate (asserted "molecule of interest") is linked to its corresponding oligonucleotide (asserted "nucleic acid moiety") by a linker containing a biotin group and a disulfide bond. Ex. 1010, 545. Kanan teaches that, "[a]fter incubations under a set of chosen reaction conditions, only pool A sequences encoding bond formation between a pool A and pool B substrate remain covalently linked to biotin." *Id.* The portion of Kanan's Figure 1.b excerpted below illustrates this process:



Id. at 546, Fig. 1.b.

Petitioner asserts that "[t]he bond formation between substrates A and B (i.e., the addition of substrate A9 to B6, [above]) would be considered the functional interaction between the substrates," and "[t]he biotin linked to the newly bound substrates is the detectable trace." Pet. 34 (citing Ex. 1005 ¶ 119). Petitioner further asserts that "Kanan discloses that after bond formation '[s]treptavidin affinity selection of the resulting solution separates biotinylated from non-biotinylated sequences,'" *Id.* at 35. Petitioner argues that "[t]he use of the biotin-streptavidin capture would allow selection of only the bond-forming combinations, i.e., demonstrate the detectable trace," which a POSA would have understood as 'selecting all cis-reactive cells exhibiting the detectable trace.'" *Id.* at 34–35 (citing Ex. 1005 ¶ 124).

In the Reply, Petitioner provides the following visualization of "[t]he associated oligonucleotides from the selected CRCs exhibiting a detectable trace":



Reply 9.

Patent Owner argues that "a critical feature in the Kanan reference . . . requires *cleaving* an oligonucleotide *before any selecting occurs*," which "results in severing one of the two alleged interactors, and thus destroying the cis-reactive cells, all before any selection occurs." PO Resp. 2, *see also id.* at 11–13, 20–28. In other words, Patent Owner argues that the above figure from Petitioner's Reply, as representative of Kanan's teaching, does not depict a selected *CRC* exhibiting a detectable trace.

We agree that Petitioner has not shown that the combination of Kanan and Neri teaches "selecting all cis-reactive cells exhibiting the detectable trace," as we have construed that phrase above. In particular, we determine that, properly construed, the limitation requires the selection of "all associations of at least two interactor moieties joined by an associated oligonucleotide and exhibiting the detectable trace." Instead, as described in Kanan, what is "selected" is the molecules of interest joined by bond formation, with only one of the molecules of interest coupled to a nucleic acid moiety. Thus, the selected molecular complex or structure does not comprise two interactors (i.e., two molecules of interest coupled to NAMs). Moreover, to the extent there are two interactors, they are joined by the bond between the molecules of interest rather than an associated oligonucleotide.

Petitioner argues in the Reply that Patent Owner improperly "[h]yper-focus[es] on a single event within Kanan's selection process," i.e., the streptavidin affinity selection. Reply 5. Petitioner argues that Kanan describes a procedure for "selecting all cis-reactive cells exhibiting the detectable trace," as that phrase is properly construed, "including *disulfide bond cleavage* and streptavidin capture." Reply Br. 7 (emphasis added). Petitioner argues that, "[w]hile PO is correct that Kanan removes pool B oligonucleotides before microarray analysis, PO is incorrect that the two interactors be unmodified or bound in proximity throughout the entire selection process." *Id.* at 10.

As an initial matter, we agree with Patent Owner that Petitioner only identified the biotin-streptavidin capture as the "selection" step in the Petition. Sur-reply 3–7; Pet. 34–35 ("The use of the biotin-streptavidin capture would allow selection of only the bond-forming combinations, i.e., demonstrate the detectable trace, which a POSA would have understood as 'selecting all cis-reactive cells exhibiting the detectable trace.'"); Ex. 1005 ¶¶ 121–124. To the extent disulfide cleavage was discussed, it was in the context of step (c), as part of the "conditions which stimulate desired functional interaction having a detectable trace." Pet. 33.

Because Petitioner may not change its theory of the case during trial from the one presented in the Petition, Petitioner must explain why the relied-upon biotin-streptavidin capture meets step 1(d), rather than attempting to point to another or an additional portion of Kanan not described in the Petition to meet this limitation. *See Intelligent Bio-Systems*, 821 F.3d at 1369 (explaining that "[i]t is of the utmost importance that petitioners in the IPR proceedings adhere to the requirement that the initial petition identify 'with particularity' the 'evidence that supports the grounds for the challenge to each claim" and holding that "the Board did not err in refusing the reply brief as improper . . . because [Appellant] relied on an entirely new rationale to explain why one of skill in the art would have been motivated to combine" the asserted prior art); *Arctic Cat Inc. v. Polaris Industries, Inc.*, 795 Fed. Appx. 827, 833 (Fed. Cir. 2019) (holding that "the Board did not misapply the law by requiring [petitioner] to prove the facts that it alleged"). We find Petitioner has not persuasively done so for the reasons discussed above.

We also agree with Patent Owner that, even if we were to consider the disulfide cleavage as part of the selection process, Petitioner has not adequately explained when and how "all cis-reactive cells exhibiting the detectable trace" were "select[ed]" at any point during the process. Ex. 1001, 41:53–54; Sur-reply 13.

As we understand Petitioner's argument, prior to the disulfide cleavage, the reaction solution contains all cis-reactive cells (i.e., A and B substrates joined by the association sequences of their respective nucleic acid moieties and any additional association resulting from polymerization). All of the cis-reactive cells comprise a biotin bound to the substrates from library B, but only some comprise the asserted functional interaction (i.e., bond formation between substrates A and B). Petitioner's argument appears to be that the disulfide cleavage and the streptavidin capture "selects" the cis-reactive cells exhibiting the detectable trace because the disulfide cleavage allows the streptavidin to capture only those biotin linked to the bound substrates, which is the alleged detectable trace.

As Patent Owner notes, however, the disulfide cleavage itself applies to all cis-reactive cells regardless of whether there has been a functional interaction between the molecules of interest. Sur-reply 13. Indeed, it is not clear that, prior to the disulfide cleavage, there are cis-reactive cells exhibiting a "detectable" trace, because, prior to the cleavage, the alleged "detectable trace" — i.e., "the biotin linked to the newly bound substrates" — cannot detectably distinguish between cis-reactive cells in which a functional reaction has occurred and those in which the reaction has not occurred. Pet. 34; Tr. 15:25–16:6 (explaining that if no disulfide cleavage takes place, all substrate pairs in Kanan, regardless of bond formation, will bind to streptavidin). However, once the disulfide bond has been cleaved, one of the molecules of interest is no longer coupled to a nucleic acid moiety, and the molecular complex or structure no longer comprises two interactor moieties, and thus do not meet the definition of a cis-reactive cell. Likewise, to the extent two interactors may be considered to be present, they are joined by the bond between the molecules of interest rather than by an associated oligonucleotide as required.

To the extent Petitioner's argument is that any step or steps enabling detection of the identification sequences in a cis-reactive cell comprising bond-forming molecules of interest reads on step 1(d), we are not persuaded for the reasons discussed above in our claim construction analysis.

b. Other arguments

Because we find Petitioner has not shown, by a preponderance of the evidence, that the combination of Kanan and Neri discloses a method of detecting functional interactions comprising a step of "selecting all cis-reactive cells exhibiting the detectable trace," we find that it has not met its burden of showing that claim 1 would have been obvious over the combination of Kanan and Neri. Accordingly, we need not reach Patent Owner's additional arguments that Kanan in view of Neri does not disclose step 1(e) ("subjecting the associated oligonucleotides from the cis-reactive cells selected in step (d) to an analysis that permits detection of the at least two identification sequence elements") or that Petitioner has not shown an ordinarily skilled artisan would have had reason to combine Kanan and Neri to arrive at the method of claim 1, with a reasonable expectation of success.

4. Conclusion as to Ground 1

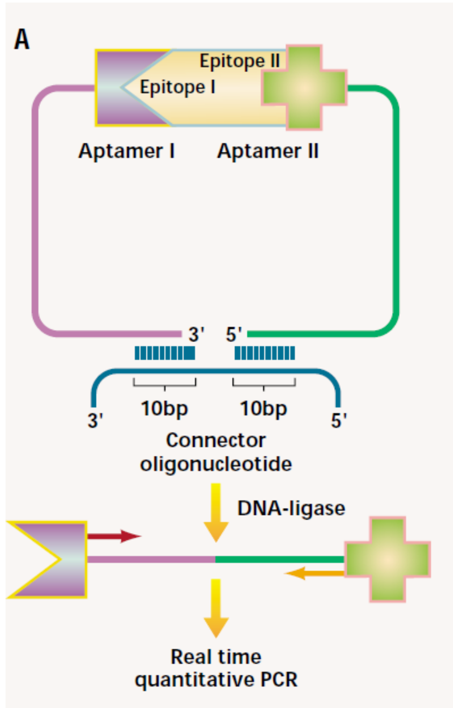
We determine that Petitioner has not shown, by a preponderance of evidence, that claim 1 would have been obvious over Kanan and Neri. Because claims 2–15 and 17–20 depend from claim 1, we find that Petitioner also has not shown these claims would have been obvious over Kanan and Neri for the same reasons discussed with respect to claim 1.

E. Ground 2: Alleged Obviousness over Kanan, Neri, and Famulok

1. Overview of Famulok (Ex. 1012)

Famulok teaches that "[t]he ability to monitor slight differences in the amounts of proteins and other biomolecules within the smallest possible detection volumes . . . is of utmost importance not only for proteomics research but also for diagnostic and technological purposes." Ex. 1012, 448.

Famulok discusses the proximity ligation method and teaches that, by "[c]ombining target recognition by two aptamers, enzymatic ligation, and PCR, the . . . method enables the detection of minute amounts of proteins." Ex. 1012, 448. In particular, Famulok discusses a test case wherein each of two DNA aptamers — nucleic acid-based receptor molecules — that binds to the homodimer of a target analyte protein includes "a different DNA-sequence extension that does not interfere with its folding and is not required for target recognition." *Id.* Famulok teaches that "[b]inding of the aptamer pair brings the ends of the oligonucleotide extensions into close spatial proximity . . . so that a 'splint' oligonucleotide can hybridize to both ends," as illustrated in Figure 1A reproduced below:



Famulok 449, Fig. 1. Figure 1(A) is a schematic that shows a pair of aptamers binding to epitopes of a target protein such that "the free ends of their sequence extensions [are brought] into close spatial proximity," which "allows the core sequence of the connector oligonucleotide to hybridize both ends, acting as a template for enzymatic ligation." *Id.* (caption).

Famulok teaches that the ends of the sequence extensions are "subsequently ligated together by [a] ligase." Ex. 1012, 448. Famulok further teaches that "[t]he ligated species can then act as a PCR template and the amplified PCR product can be monitored and quantified under real-time PCR conditions, whereas no signal is obtained with unligated probes." *Id.* Famulok teaches that "[t]he sensitivity that can be achieved with the method is remarkable." *Id.* Famulok further teaches that "there is no reason that the approach should be limited to aptameric receptors" and that "[p]roximity ligation will likely work for antibody probes that are linked to a DNA sequence." *Id.* at 449.

2. Analysis of Claim 16

Claim 16 depends from claim 1 and further recites "wherein association [between at least two nucleic acid moieties in the cis-reactive cell] is via ligation." Ex. 1001, 41:37–57 (claim 1), 43:20–22 (claim 16). Petitioner asserts that Famulok discloses this additional limitation because it teaches "associating the oligonucleotides of [its] aptamers by connecting the two oligonucleotides to a third 'splint' oligonucleotide that is incorporated into double-stranded DNA by DNA ligase." Pet. 42.

We agree with Patent Owner that the Petition does not explain how Famulok cures the deficiency of Neri and Kanan with respect to step 1(d), "selecting all cis-reactive cells exhibiting the detectable trace." PO Resp, 49–50. Accordingly, Petitioner also fails to establish that claim 16, which depends from claim 1, is unpatentable over Kanan, Neri, and Famulok, for the same reasons discussed above with respect to Ground 1.

3. Conclusion as to Ground 2

We determine that Petitioner has not shown, by a preponderance of evidence, that claim 16 would have been obvious over Kanan, Neri, and Famulok.

F. Grounds 3 and 4: Alleged Obviousness over Baez and Landegren or Baez, Landegren, and Freskard

1. Analysis of claims 1–8, 11, 12, 14, and 15

As noted above, Petitioner states that, "[f]or Grounds 3 and 4, the claim terms are given meaning based on PO's application of the claims in related litigation with Petitioner." Pet. 12–13 (citing Exs. 1007, 1008, 1015). Petitioner asserts that, although "[t]here are no prior claim construction rulings on the ’848 patent . . . , [Patent Owner] filed a Complaint (Ex. 1007) and later provided an infringement chart (Ex. 1008) reflecting [Patent Owner’s] application of the claim terms.” *Id.* at 13 (footnote omitted).

Likewise, Dr. Nelson explained that, for these grounds, he “relied on Patent Owner’s positions provided in its complaint, infringement chart, and opposition to [Petitioner’s] motion to dismiss and [applies] those positions to how a POSA would have understood the claims as if Patent Owner’s positions were correct at the time of the application.” Ex. 1005 ¶ 210. Dr. Nelson emphasized that “[his] analysis for Grounds 3 and 4 do[es] not reflect [his] opinion on what a POSA’s ordinary understanding of the scope of the ’848 Patent [would have been].” *Id.*

Finally, Petitioner’s counsel agreed at oral argument that, if the claims were construed as a person of ordinary skill would have understood them, Baez and Landegren would not render the claims obvious. Tr. 21:6–16.

As noted above, however, in AIA proceedings we give claims their ordinary and customary meaning, as would have been understood by a person of ordinary skill in the art, at the time of the invention, in light of the language of the claims, the specification, and the prosecution history of record. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–19 (Fed. Cir. 2005) (en banc); 37 C.F.R. § 42.100(b). Therefore, because Petitioner has not offered evidence that the combination of Baez and Landegren renders obvious claims 1–8, 11, and 12 as properly construed, we find that Petitioner has not demonstrated by a preponderance of the evidence that claims 1–8, 11, and 12 would have been obvious over the combination of Baez and Landegren. For the same reasons, we find Petitioner has not demonstrated by a preponderance of evidence that claims 14 and 15 would have been obvious over the combination of Baez, Landegren, and Freskard.

2. Conclusion as to Grounds 3 and 4

We determine that Petitioner has not shown, by a preponderance of evidence, that claims 1–8, 11, and 12 would have been obvious over Baez and Landegren. We likewise determine that Petitioner has not shown, by a preponderance of evidence, that claims 14 and 15 would have been obvious over Baez, Landegren, and Freskard.

III. PATENT OWNER'S MOTION TO EXCLUDE EVIDENCE

Patent Owner moved to exclude Exhibit 1008, on the ground that it contains settlement statements and that Rule 408 of the Federal Rule of Evidence prohibits Petitioner from using the statements to impeach or contradict Patent Owner. Mot. 5. Because we do not rely in our analysis on this exhibit, we dismiss the Motion to Exclude as moot.

IV. CONCLUSION

Based on the information presented, we conclude that Petitioner has not demonstrated by a preponderance of evidence that claims 1–20 are unpatentable. We dismiss Patent Owner's Motion to Exclude as moot.

In summary:

Claims	**35 U.S.C. §**	**Reference(s)/Basis**	**Claims Shown Unpatentable**	**Claims Not Shown Unpatentable**
1–15, 17–20	103(a)	Kanan, Neri		1–15, 17–20
16	103(a)	Kanan, Neri, Famulok		16
1–8, 11, 12	103(a)	Baez, Landegren		1–8, 11, 12
14, 15	103(a)	Baez, Landegren, Freskard		14, 15
Overall Outcome				1–20

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has not demonstrated by a preponderance of the evidence that claims 1–20 of U.S. Patent No. 7,883,848 are unpatentable; and

FURTHER ORDERED that Patent Owner's Motion to Exclude is *dismissed* as moot; and

FURTHER ORDERED that, because this is a final written decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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